

Multidimensional Brownian Motion

The point of this chapter is to establish the connection between Brownian motion and various parabolic and elliptic partial differential equations. The first section uses martingales to do computations for multidimensional Brownian motion.

9.1 Martingales

Let $\Delta f = \sum_{i=1}^d \partial^2 f / \partial x_i^2$ be the Laplacian of f . Combining Ito's formula (7.6.9) with Theorem 7.6.4:

Theorem 9.1.1 *Suppose $v \in C^2$, i.e., all first- and second-order partial derivatives exist and are continuous, and $E \int_0^t |\nabla v(B_s)|^2 ds < \infty$ for all t . Then*

$$v(B_t) - \int_0^t \frac{1}{2} \Delta v(B_s) ds \quad \text{is a continuous martingale.}$$

Let $S_r = \inf\{t : |B_t| = r\}$, $r < R$, and $\tau = S_r \wedge S_R$. The first detail is to note that if $|x| < R$, then $P_x(S_R < \infty) = 1$. Once we know this we can conclude

Theorem 9.1.2 *If $|x| < R$, then $E_x S_R = (R^2 - |x|^2)/d$.*

Proof It follows from Theorem 7.5.4 that $|B_t|^2 - dt = \sum_{i=1}^d (B_t^i)^2 - t$ is a martingale. Theorem 7.5.1 implies $|x|^2 = E|B_{S_R \wedge t}|^2 - dE(S_R \wedge t)$. Letting $t \rightarrow \infty$ gives the desired result. $\square$

To further investigate Brownian motion, we introduce two new functions:

$$\varphi(x) = \begin{cases} \log |x| & d = 2 \\ |x|^{2-d} & d \geq 3 \end{cases}$$

We leave it to the reader to check that in each case $\Delta \varphi = 0$.

Lemma 9.1.3 $\varphi(x) = E_x \varphi(B_\tau)$.

Proof Define $\psi(x) = g(|x|)$ to be C^2 and have compact support, and have $\psi(x) = \varphi(x)$ when $r < |x| < R$. Theorem 9.1.1 implies that $\psi(x) = E_x \psi(B_{t \wedge \tau})$. Letting $t \rightarrow \infty$ now gives the desired result. $\square$

Lemma 9.1.3 implies that

$$\varphi(x) = \varphi(r)P_x(S_r < S_R) + \varphi(R)(1 - P_x(S_r < S_R))$$

where $\varphi(r)$ is short for the value of $\varphi(x)$ on $\{x : |x| = r\}$. Solving now gives

$$P_x(S_r < S_R) = \frac{\varphi(R) - \varphi(x)}{\varphi(R) - \varphi(r)} \quad (9.1.1)$$

In $d = 2$, the last formula says

$$P_x(S_r < S_R) = \frac{\log R - \log |x|}{\log R - \log r} \quad (9.1.2)$$

If we fix r and let $R \rightarrow \infty$ in (9.1.2), the right-hand side goes to 1. So

$$P_x(S_r < \infty) = 1 \quad \text{for any } x \text{ and any } r > 0$$

It follows that two-dimensional Brownian motion is **recurrent** in the sense that if G is any open set, then $P_x(B_t \in G \text{ i.o.}) \equiv 1$.

If we fix R , let $r \rightarrow 0$ in (9.1.2), and let $S_0 = \inf\{t > 0 : B_t = 0\}$, then for $x \neq 0$

$$P_x(S_0 < S_R) \leq \lim_{r \rightarrow 0} P_x(S_r < S_R) = 0$$

Since this holds for all R and since the continuity of Brownian paths implies $S_R \uparrow \infty$ as $R \uparrow \infty$, we have $P_x(S_0 < \infty) = 0$ for all $x \neq 0$. To extend the last result to $x = 0$, we note that the Markov property implies

$$P_0(B_t = 0 \text{ for some } t \geq \epsilon) = E_0[P_{B_\epsilon}(T_0 < \infty)] = 0$$

for all $\epsilon > 0$, so $P_0(B_t = 0 \text{ for some } t > 0) = 0$, and thanks to our definition of $S_0 = \inf\{t > 0 : B_t = 0\}$, we have

$$P_x(S_0 < \infty) = 0 \quad \text{for all } x \quad (9.1.3)$$

Thus, in $d \geq 2$ Brownian motion will not hit 0 at a positive time even if it starts there. As a corollary we see that Brownian motion in $d = 3$ will not hit a straight line.

For $d \geq 3$, formula (9.1.1) says

$$P_x(S_r < S_R) = \frac{R^{2-d} - |x|^{2-d}}{R^{2-d} - r^{2-d}} \quad (9.1.4)$$

There is no point in fixing R and letting $r \rightarrow 0$, here. The fact that two-dimensional Brownian motion does not hit 0 implies that three-dimensional Brownian motion does not hit 0 and indeed will not hit the line $\{x : x_1 = x_2 = 0\}$. If we fix r and let $R \rightarrow \infty$ in (9.1.4), we get

$$P_x(S_r < \infty) = (r/|x|)^{d-2} < 1 \quad \text{if } |x| > r \quad (9.1.5)$$

From the last result it follows easily that for $d \geq 3$, Brownian motion is **transient**, i.e., it does not return infinitely often to any bounded set.

Theorem 9.1.4 *As $t \rightarrow \infty$, $|B_t| \rightarrow \infty$ a.s.*

Proof Let $A_n = \{|B_t| > n^{1-\epsilon} \text{ for all } t \geq S_n\}$. The strong Markov property implies

$$P_x(A_n^c) = E_x(P_{B(S_n)}(S_{n^{1-\epsilon}} < \infty)) = (n^{1-\epsilon}/n)^{d-2} \rightarrow 0$$

as $n \rightarrow \infty$. Now $\limsup A_n = \bigcap_{N=1}^{\infty} \bigcup_{n=N}^{\infty} A_n$ has

$$P(\limsup A_n) \geq \limsup P(A_n) = 1$$

So infinitely often the Brownian path never returns to $\{x : |x| \leq n^{1-\epsilon}\}$ after time S_n and this implies the desired result. $\square$

The scaling relation (7.1.1) implies that $S_n =_d n^2 S_1$, so the proof of Theorem 9.1.4 suggests that

$$|B_t|/t^{(1-\epsilon)/2} \rightarrow \infty$$

Dvoretzky and Erdős (1951) have proved the following result about how fast Brownian motion goes to ∞ in $d \geq 3$.

Theorem 9.1.5 *Suppose $g(t)$ is positive and decreasing. Then*

$$P_0(|B_t| \leq g(t)\sqrt{t} \text{ i.o. as } t \uparrow \infty) = 1 \text{ or } 0$$

according as $\int_0^\infty g(t)^{d-2}/t \, dt = \infty$ or $< \infty$.

Here the absence of the lower limit implies that we are only concerned with the behavior of the integral “near ∞ .” A little calculus shows that

$$\int_0^\infty t^{-1} \log^{-\alpha} t \, dt = \infty \text{ or } < \infty$$

according as $\alpha \leq 1$ or $\alpha > 1$, so B_t goes to ∞ faster than

$$t^{1/2}(\log t)^{-\alpha/(d-2)} \quad \text{for any } \alpha > 1.$$

Exercises

9.1.1 Combine scaling with the 0-1 law to show that

$$\liminf_{t \rightarrow \infty} |B_t|/t^{1/2} = 0 \quad \text{a.s.}$$

9.1.2 Suppose $f \in C^2$ is nonnegative and superharmonic. Show that in dimensions 1 and 2, f must be constant. Give an example of the form $f(x) = g(|x|)$ in $d \geq 3$.

9.1.3 Suppose $f \in C^2$ is bounded and harmonic. Then f must be constant (in any dimension).

9.2 Heat Equation

In this section, we will consider the following equation:

(a) $u \in C^{1,2}$ and $u_t = \frac{1}{2} \Delta u$ in $(0, \infty) \times \mathbf{R}^d$.

(b) u is continuous at each point of $\{0\} \times \mathbf{R}^d$ and $u(0, x) = f(x)$, where f is bounded and continuous.

This equation derives its name from the fact that if the units of measurement are chosen suitably, then the solution $u(t, x)$ gives the temperature at the point $x \in \mathbf{R}^d$ at time t when the temperature profile at time 0 is given by $f(x)$.

Obviously, (b) cannot hold unless f is continuous. The results below can be generalized to unbounded f . We assume f is bounded to keep things simple. In this and the next five sections the assumptions stated at the beginning are in force throughout the section.

The first step in solving this equation, as it will be six times in the following, is to find a local martingale M_s , $s \in [0, t]$, i.e., a process in which there is an increasing sequence of stopping times so that $M(s \wedge T_n)$ is a martingale and $P(T_n < t) \rightarrow 0$.

Theorem 9.2.1 *If u satisfies (a), $M_s = u(t - s, B_s)$ is a local martingale on $[0, t]$.*

Proof Applying Itô's formula, (7.6.9) gives

$$\begin{aligned} u(t - s, B_s) - u(t, B_0) &= \int_0^s -u_t(t - r, B_r) dr \\ &+ \int_0^s \nabla u(t - r, B_r) \cdot dB_r + \frac{1}{2} \int_0^s \Delta u(t - r, B_r) dr \end{aligned}$$

The result follows, since $-u_t + \frac{1}{2}\Delta u = 0$ and the second term on the right-hand side is a local martingale by Theorem 7.6.4. Here we need the stopping times to make sure the integrability condition holds. $\square$

Our next step is to prove a uniqueness theorem.

Theorem 9.2.2 *If there is a solution that is bounded, then it must be*

$$v(t, x) \equiv E_x f(B_t)$$

Here $\equiv$ means that the last equation defines v . We will always use u for a generic solution of the equation and v for our special solution.

Proof If we now assume that u is bounded, then $M_s, 0 \leq s < t$, is a bounded martingale. The martingale convergence theorem implies that

$$M_t \equiv \lim_{s \uparrow t} M_s \quad \text{exists a.s.}$$

If u satisfies (b), this limit must be $f(B_t)$. Since M_s is uniformly integrable, it follows that $u(t, x) = E_x M_0 = E_x M_t = v(t, x)$. $\square$

Now that Theorem 9.2.2 has told us what the solution must be, the next logical step is to find conditions under which v is a solution. Here and in the other five examples it is easy to show that if v is smooth enough, then it is a solution.

Theorem 9.2.3 *If $v \in C^{1,2}$, then it satisfies (a).*

Proof If we think of running space time Brownian motion $(t - s, B_s)$ to construct the solution, then we see

$$v(t + h, x) = E_x v(t, B_h)$$

Subtracting $v(t, x)$ from each side and using Taylor series

$$v(t+h, x) - v(t, x) = E_x \left(\sum_{i=1}^d \frac{\partial v}{\partial x_i}(t, x)(B_i(h) - x_i) + \frac{1}{2} \sum_{1 \leq i, j \leq d} \frac{\partial^2 v}{\partial x_i \partial x_j}(t, x)(B_i(h) - x_i)(B_j(h) - x_j) \right) + o(h)$$

Since for $i \neq j$, $B_i(t)$, $B_i^2(t) - t$, and $B_i(t)B_j(t)$ are martingales

$$v(t+h, x) - v(t, x) = \frac{1}{2} \sum_{i=1}^d \frac{\partial^2 v}{\partial x_i^2}(t, x)h + o(h)$$

Dividing by h and letting $h \rightarrow 0$ gives the desired result. $\square$

Theorem 9.2.4 v satisfies (b).

Proof $(B_t - B_0) \stackrel{d}{=} t^{1/2}\chi$, where χ has a normal distribution with mean 0 and variance 1, so if $t_n \rightarrow 0$ and $x_n \rightarrow x$, the bounded convergence theorem implies that

$$v(t_n, x_n) = Ef(x_n + t_n^{1/2}\chi) \rightarrow f(x)$$

which proves the desired result. $\square$

The final step in showing that v is a solution is to find conditions that guarantee that it is smooth.

Theorem 9.2.5 $v \in C^{1,2}$ and hence satisfies (a).

Proof Our solution is

$$v(t, x) = E_x f(B_t) = \int (2\pi t)^{-d/2} e^{-|x-y|^2/2t} f(y) dy$$

Using calculus and some basic facts from analysis one can prove the desired result. The details are not exciting and are left to the reader. $\square$

9.3 Inhomogeneous Heat Equation

In this section, we will consider what happens when we add a bounded continuous function $g(x)$ to the heat equation. That is, we will study

(a) $u \in C^{1,2}$ and $u_t = \frac{1}{2}\Delta u + g$ in $(0, \infty) \times \mathbf{R}^d$

(b) u is continuous at each point of $\{0\} \times \mathbf{R}^d$ and $u(0, x) = f(x)$ is bounded and continuous.

We observed in the previous section that (b) cannot hold unless f is continuous. Here $g = u_t - (1/2)\Delta u$, so the equation in (a) cannot hold with $u \in C^{1,2}$ unless g is continuous. Our first step in treating the new equation is to observe that if u_1 is a solution of the equation with $f = f_0$ and $g = 0$ which we studied in the last section, and u_2 is a solution of the equation with $f = 0$ and $g = g_0$, then $u_1 + u_2$ is a solution of the equation with $f = f_0$ and $g = g_0$, so we can restrict our attention to the case $f \equiv 0$.

Having made this simplification, we will now study the previous equation by following the procedure used in the last section.

Theorem 9.3.1 *If u satisfies (a), then*

$$M_s = u(t - s, B_s) + \int_0^s g(B_r) dr$$

is a local martingale on $[0, t]$.

Proof Applying Itô's formula as in the proof of Theorem 9.2.1 gives

$$\begin{aligned} u(t - s, B_s) - u(t, B_0) &= \int_0^s (-u_t + \frac{1}{2} \Delta u)(t - r, B_r) dr \\ &\quad + \int_0^s \nabla u(t - r, B_r) \cdot dB_r \end{aligned}$$

which proves the result, since $-u_t + \frac{1}{2} \Delta u = -g$ and the second term on the right-hand side is a local martingale by Theorem 7.6.4. $\square$

Again the next step is a uniqueness result.

Theorem 9.3.2 *If there is a solution that is bounded on $[0, T] \times \mathbf{R}^d$ for any $T < \infty$, it must be*

$$v(t, x) \equiv E_x \left(\int_0^t g(B_s) ds \right)$$

Proof Under the assumptions on g and u , $M_s, 0 \leq s < t$, defined in Theorem 9.3.1 is a bounded martingale and $u(0, x) \equiv 0$, so

$$M_t \equiv \lim_{s \uparrow t} M_s = \int_0^t g(B_s) ds$$

Since M_s is uniformly integrable, $u(t, x) = E_x M_0 = E_x M_t = v(t, x)$. $\square$

Again, it is easy to show that if v is smooth enough, it is a solution.

Theorem 9.3.3 *If $v \in C^{1,2}$, then it satisfies (a) in $(0, \infty) \times \mathbf{R}^d$.*

Proof If we think of running space time Brownian motion $(t - s, B_s)$ to construct the solution, then we see

$$v(t + h, x) = E_x \left(v(t, B_h) + \int_0^h g(B_s) ds \right)$$

Subtracting $v(t, x)$ from each side and using Taylor series

$$\begin{aligned} v(t + h, x) - v(t, x) &= E_x \left(\sum_{i=1}^d \frac{\partial v}{\partial x_i}(t, x) (B_i(h) - x_i) \right. \\ &\quad + \frac{1}{2} \sum_{1 \leq i, j \leq d} \frac{\partial^2 v}{\partial x_i \partial x_j}(t, x) (B_i(h) - x_i) (B_j(h) - x_j) \\ &\quad \left. + \int_0^h g(B_s) ds \right) + o(h) \end{aligned}$$

Since for $i \neq j$, $B_i(t)$, $B_i^2(t) - t$, and $B_i(t)B_j(t)$ are martingales

$$v(t+h, x) - v(t, x) = \frac{1}{2} \sum_{i=1}^d \frac{\partial^2 v}{\partial x_i^2}(t, x) h + E_x \int_0^h g(B_s) ds + o(h)$$

Dividing by h and letting $h \rightarrow 0$ gives $v_t = \frac{1}{2} \Delta V + g(x)$, which is the desired result. $\square$

Theorem 9.3.4 v satisfies (b).

Proof If $|g| \leq M$, then as $t \rightarrow 0$

$$|v(t, x)| \leq E_x \int_0^t |g(B_s)| ds \leq Mt \rightarrow 0$$

which proves the desired result. $\square$

The final step in showing that v is a solution is to check that $v \in C^{1,2}$. The calculations are messy, so we just state the result. Here and in what follows when we say a function is Hölder continuous, that means $|g(x) - g(y)| \leq C|x - y|^\alpha$ for some $\alpha > 0$.

Theorem 9.3.5 If g is Hölder continuous, then $v \in C^{1,2}$.

The problem is that $v(t, x) = \int G_t(x, y)g(y) dy$, where

$$G_t(x, y) = \int_0^t (2\pi s)^{-d/2} e^{-|y-x|^2/2s} ds$$

Looking at the y with $|y - x|^2 \leq s$ leads to a lower bound

$$G_t(x, y) \geq C \int_{|y-x|^2}^t s^{-d/2} ds = O(|y - x|^{2-d})$$

From this it follows that first derivative blows up like δ^{1-d} and the second like is δ^{-d} , which is not integrable near 0. However, if g is Hölder continuity of g , its smoothness can save the day. For details, see pp 12–13 of Friedman (1964).

9.4 Feynman-Kac Formula

In this section, we will consider what happens when we add cu to the right-hand side of the heat equation, where $c(x)$ is bounded and continuous. That is, we will study

(a) $u \in C^{1,2}$ and $u_t = \frac{1}{2} \Delta u + cu$ in $(0, \infty) \times \mathbf{R}^d$.

(b) u is continuous at each point of $\{0\} \times \mathbf{R}^d$ and $u(0, x) = f(x)$, where f is bounded and continuous.

If $c(x) \leq 0$, then this equation describes heat flow with cooling. That is, $u(t, x)$ gives the temperature at the point $x \in \mathbf{R}^d$ at time t , when the heat at x at time t dissipates at the rate $-c(x)$.

The first step, as usual, is to find a local martingale.

Theorem 9.4.1 Let $c_s = \int_0^s c(B_r) dr$. If u satisfies (a), then

$$M_s = u(t - s, B_s) \exp(c_s)$$

is a local martingale on $[0, t]$

Proof The first step in doing this is to generalize our third version of Itô's formula

$$\begin{aligned} & u(t-s, B_s) \exp(c_s) - u(t, B_0) \\ &= \int_0^s -u_t(t-r, B_r) \exp(c_r) dr + \int_0^s \exp(c_r) \nabla u(t-r, B_r) \cdot dB_r \\ &+ \int_0^s u(t-r, B_r) c(B_r) \exp(c_r) dr + \frac{1}{2} \int_0^s \Delta u(t-r, B_r) \exp(c_r) dr \end{aligned} \quad (9.4.1)$$

Proof of (9.4.1). As in the three previous proofs of Itô's formula, we subdivide the interval $[0, s]$ with $s_i^n = si/2^n$. We write the difference as

$$\begin{aligned} & [u(t-s_{i+1}^n, B(s_{i+1}^n)) - u(t-s_i^n, B(s_i^n))] \exp(c(s_{i+1}^n)) \\ &+ u(t-s_i^n, B(s_i^n)) [\exp(c(s_{i+1}^n)) - \exp(c(s_i^n))] \end{aligned}$$

The limit of the first line gives the first, second, and fourth terms. Using

$$\exp(c(s_{i+1}^n)) - \exp(c(s_i^n)) = \exp(c(s_i^n)) \left\{ \exp \left(\int_{s_i^n}^{s_{i+1}^n} c(B_r) dr \right) - 1 \right\}$$

we see that the limit of the second line gives the third term. $\square$

Rearranging, the right-hand side of (9.4.1) is

$$\begin{aligned} &= \int_0^s \left(-u_t + cu + \frac{1}{2} \Delta u \right) (t-r, B_r) \exp(c_r) dr \\ &+ \int_0^s \exp(c_r) \nabla u(t-r, B_r) \cdot dB_r \end{aligned}$$

which proves the desired result, since $-u_t + cu + \frac{1}{2} \Delta u = 0$, so using Theorem 7.6.4 the second term is a local martingale. $\square$

The next step, again, is a uniqueness result.

Theorem 9.4.2 *If there is a solution that is bounded on $[0, T] \times \mathbf{R}^d$ for any $T < \infty$, it must be*

$$v(t, x) \equiv E_x \{ f(B_t) \exp(c_t') \} \quad (9.4.2)$$

Proof Under our assumptions on c and u , M_s , $0 \leq s < t$, is a bounded martingale and $M_t \equiv \lim_{s \rightarrow t} M_s = f(B_t) \exp(c_t)$. Since M_s is uniformly integrable, it follows that

$$u(t, x) = E_x M_0 = E_x M_t = v(t, x)$$

which proves the desired result. $\square$

As before, it is easy to show that if v is smooth enough, it is a solution.

Theorem 9.4.3 *If $v \in C^{1,2}$, then it satisfies (a).*

Proof When we write the solution in (9.4.2), we think about running space-time Brownian motion $(t-s, B_s)$ from time 0 to t , so we have

$$v(t+h, x) = E_x [v(t, B_h) \exp(c_h)]$$

Subtracting $v(t, x)$ from each side

$$v(t + h, x) - v(t, x) = E_x[v(t, B_h) \exp(c_h) - v(t, x)]$$

Adding and subtracting $v(t, B_h)$ and the right-hand side and using Taylor's theorem

$$\begin{aligned} v(t, B_h) \exp(c_h) - v(t, x) &\approx \sum_{i=1}^d \frac{\partial v}{\partial x_i}(t, x)(B_h^i - x) \\ &+ \frac{1}{2} \sum_{1 \leq i, j \leq d} \frac{\partial^2 v}{\partial x_i \partial x_j}(t, x)(B_h^i - x)(B_h^j - x) + v(t, B_h)(\exp(c_h) - 1) + o(h) \end{aligned}$$

Since for $i \neq j$, $W_i(t)$, $W_i^2(t) = t$ and $W_i(t)W_j(t)$ are martingales.

$$E_x[v(t, B_h) \exp(c_h) - v(t, x)] \approx \sum_{i=1}^d \frac{\partial^2 v}{\partial x_i^2} h + E_x[v(t, B_h)(\exp(c_h) - 1)] + o(h)$$

Dividing each side by h and letting $h \rightarrow 0$, we get

$$\frac{\partial v}{\partial t} = \frac{1}{2} \Delta v(t, x) + c(x)v(t, x)$$

which proves the desired result. $\square$

Theorem 9.4.4 v satisfies (b).

Proof If $|c| \leq M$, then $e^{-Mt} \leq \exp(c_t^i) \leq e^{Mt}$, so $\exp(c_t^i) \rightarrow 1$ as $t \rightarrow 0$. Letting $\|f\|_\infty = \sup_x |f(x)|$ this result implies

$$|E_x \exp(c_t^i) f(B_t) - E_x f(B_t)| \leq \|f\|_\infty |E_x \exp(c_t^i) - 1| \rightarrow 0$$

Theorem 9.2.4 implies that $(t, x) \rightarrow E_x f(B_t)$ is continuous at each point of $\{0\} \times \mathbf{R}^d$ and the desired result follows. $\square$

This brings us to the problem of determining when v is smooth enough to be a solution.

Theorem 9.4.5 If f and c Hölder continuous, then $v \in C^{1,2}$ and, hence, satisfies (a).

To solve the problem in this case, we use a trick to reduce our result to the previous case. We begin by observing that

$$c_s = \int_0^s c(B_r) dr$$

is continuous and locally of bounded variation. So using the ordinary change of variables formula, if $h \in C^1$, then

$$h(c_t) - h(c_0) = \int_0^t h'(c_s) c(B_s) ds$$

Taking $h(x) = e^{-x}$, we have

$$\exp(-c_t) - 1 = - \int_0^t \exp(-c_s) c(B_s) ds$$

Multiplying by $-\exp(c_t)$ gives

$$\exp(c_t) - 1 = \int_0^t c(B_s) \exp(c_t - c_s) ds$$

Plugging in the definitions of c_t^t and c_s^t , we have

$$\exp\left(\int_0^t c(B_r) dr\right) = 1 + \int_0^t c(B_s) \exp\left(\int_s^t c(B_r) dr\right) ds$$

Multiplying by $f(B_t)$, taking expected values, and using Fubini's theorem, which is justified since everything is bounded, gives

$$v(t, x) = E_x f(B_t) + \int_0^t E_x \left\{ c(B_s) \exp\left(\int_s^t c(B_r) dr\right) f(B_t) \right\} ds$$

Conditioning on $\mathcal{F}_s$, noticing $c(B_s) \in \mathcal{F}_s$, and using the Markov property

$$E_x \left(c(B_s) \exp\left(\int_s^t c(B_r) dr\right) f(B_t) \middle| \mathcal{F}_s \right) = c(B_s) v(t-s, B_s)$$

Taking the expected value of the last equation and plugging into the previous one, we have

$$\begin{aligned} v(t, x) &= E_x f(B_t) + \int_0^t E_x \{ c(B_s) v(t-s, B_s) \} ds \\ &\equiv v_1(t, x) + v_2(t, x) \end{aligned} \tag{9.4.3}$$

The first term on the right, $v_1(t, x)$, is $C^{1,2}$ by Theorem 9.2.3. The second term, $v_2(t, x)$, is of the form considered in the last section with $g(r, x) = c(x)v(r, x)$. If we start with the trivial observation that if c and f are bounded, then v is bounded on $[0, N] \times \mathbf{R}^d$, and apply (2.6b), we see that

$$|v_2(t, x) - v_2(t, y)| \leq C_N |x - y| \quad \text{whenever } t \leq N$$

To get a similar estimate for v_1 let $\bar{B}_t$ be a Brownian motion starting at 0 and observe that since f is Hölder continuous

$$\begin{aligned} |v_1(t, x) - v_1(t, y)| &= |E\{f(x + \bar{B}_t) - f(y + \bar{B}_t)\}| \\ &\leq E|f(x + \bar{B}_t) - f(y + \bar{B}_t)| \leq C|x - y|^\alpha \end{aligned}$$

Combining the last two estimates, we see that $v(r, x)$ is Hölder continuous locally in t . Since c and v are bounded and we have supposed that c, x is Hölder continuous, the triangle inequality implies $g(r, x)$ is Hölder continuous locally in t . The result then follows from smoothness results for the inhomogeneous heat equation.

9.5 Dirichlet Problem

The developments here and in Section 9.7 follow Port and Stone (1978). In this section, we will consider the Dirichlet problem. That is, given an open set G , we will seek a function that satisfies

(a) $u \in C^2$ and $\Delta u = 0$ in G .

(b) At each point of ∂G , u is continuous and $u = f$, a bounded function.

Physically, u is an equilibrium temperature distribution when ∂G is held at a fixed temperature profile f .

Let $\tau = \inf\{t > 0 : B_t \notin G\}$. We assume throughout this section that

$$P_x(\tau < \infty) = 1 \quad \text{for all } x \in G. \quad (9.5.1)$$

If u satisfies (a), then $M_t = u(B_t)$ is a local martingale on $[0, \tau)$. That is, there is an increasing sequence of stopping times T_n so that $u(B(t \wedge T_n))$ is a martingale. To turn a local martingale into an honest martingale, we introduce a function $\gamma : [0, \infty) \rightarrow [0, \tau)$ stays constant for one unit of time when its value reaches T_n . That is, $\gamma(t) = t$ for $0 \leq t \leq T_1$, $\gamma(t) = T_1$ for $T_1 \leq t \leq T_1 + 1$, and in general

$$\begin{aligned} \gamma(t) &= t - k \text{ for } T_k + k \leq t \leq T_{k+1} + k \\ \gamma(t) &= T_{k+1} \text{ for } T_{k+1} + k \leq t \leq T_{k+1} + k + 1 \end{aligned}$$

Since $\gamma(t)$ is a stopping time and $\gamma(n) \leq T_n$.

Lemma 9.5.1 $M_t = u(B(\gamma(t)))$ is a martingale.

Theorem 9.5.2 If there is a solution of the Dirichlet problem that is bounded, it must be

$$v(x) \equiv E_x f(B_\tau)$$

Proof Since u is bounded, M_t is a bounded martingale, and hence $\lim_{t \rightarrow \infty} M_t$ exists. The boundary condition implies $M_\infty = f(B_\tau)$. Since the martingale is uniformly integrable, $u(x) = E_x M_\infty = E_x f(B_\tau)$. $\square$

Theorem 9.5.3 If $v \in C^2$, then it satisfies (a).

Proof We learned this argument, which avoids the use of stochastic calculus, from Liggett (2010). Let $x \in D$ and $B = B(x, r) \subset D$. If τ_B is the exit time from B , then the strong Markov property implies

$$v(x) = E_x v(B(\tau_B)).$$

Using Taylor's theorem

$$\begin{aligned} E_x v(B(\tau_B)) - v(x) &= E_x \sum_{i=1}^d \frac{\partial v}{\partial x_i}(x) [B_i(\tau_B) - x_i] \\ &\quad + \frac{1}{2} E_x \sum_{1 \leq i, j \leq d} \frac{\partial^2 v}{\partial x_i \partial x_j}(x) [B_i(\tau_B) - x_i] [B_j(\tau_B) - x_j] + o(r^2) \end{aligned}$$

Since for $i \neq j$, $B_i(t)$, $B_i^2(t) - t$ and $B_i(t)B_j(t)$ are martingales, we have

$$\begin{aligned} E_x [B_i(\tau_B) - x_i] &= 0 & E_x [B_i(\tau_B) - x_i] [B_j(\tau_B) - x_j] &= 0 \quad \text{when } i \neq j \\ E_x [B_i(\tau_B) - x_i]^2 &= E_0 [B_i^2(\tau_B)] = E_x \tau_B \end{aligned}$$

Using these results in the previous formula

$$0 = E_x v(B(\tau_B)) - v(x) = \frac{1}{2} \Delta v(x) E_x \tau_B + o(r^2)$$

which proves the desired result. $\square$

Theorem 9.5.4 $v \in C^\infty$.

Proof Let $x \in D$ and $B = B(x, r) \subset D$. If τ_B is the exit time from B , then the strong Markov property implies

$$v(x) = E_x v(B(\tau_B)) = \int_{\partial D(x, r)} \bar{v}(y) \pi(dy).$$

where π is surface measure on the boundary $\partial D(x, r)$ normalized to be a probability measure.

The last result is the **averaging property** of harmonic functions. Now a simple analytical argument takes over to show $v \in C^\infty$. To prove this, we let ψ be a nonnegative infinitely differentiable function that vanishes on $[\delta^2, \infty)$ but is not $\equiv 0$. It is routine to show that

$$h(x) = \int_{D(x, \delta)} \psi(|y - x|^2) v(y) dy \in C^\infty$$

Changing to spherical coordinates and using the averaging property, it is easy to see that $h(x)$ is a constant multiple of $v(x)$ so $v \in C^\infty$. $\square$

As the next example shows, v does not always satisfy (b).

Example 9.5.5 (Punctured Disc) Let $d = 3$ and let $G = D - K$, where $D = \{x : |x| < 1\}$, and $K = \{x : x_1 = x_2 = 0, x_3 \geq 0\}$. Our uniqueness result and the fact that three-dimensional Brownian motion never hits a line implies that the solution must be $v(x) = E_x f(B_{\tau(D)})$, where $\tau(D)$ is the exit time from D . If we let f be 0 on the boundary of D and $1 - x_3$ on K , then the solution $v \equiv 0$ in G and boundary condition will not be satisfied.

Example 9.5.6 (Lebesgue's Thorn) Let $d \geq 3$ and let

$$G = (-1, 1)^d - \cup_{1 \leq n \leq \infty} \{[2^{-n}, 2^{-n+1}] \times [-a_n, a_n]^{d-1}\}$$

where the $n = \infty$ term is the single point $\{0\}$. We will now show that if $a_n \downarrow 0$ sufficiently fast, then $P_0(\tau = 0) = 0$. From this, it will follow that if we define f to be 0 on the boundary of $(-1, 1)^d$ and $1 - x_1$ on the rest of the boundary, $E_0 f(B_\tau) < 1$ and the boundary condition is not satisfied.

Proof $P_0((B_t^2, B_t^3) = (0, 0) \text{ for some } t > 0) = 0$, so with probability 1, a Brownian motion B_t starting at 0 will not hit

$$I_n = \{x : x_1 \in [2^{-n}, 2^{-n+1}], x_2 = x_3 = \dots = x_d = 0\}$$

Since B_t is transient in $d \geq 3$, it follows that for a.e. ω the distance between $\{B_s : 0 \leq s < \infty\}$ and I_n is positive. From the last observation, it follows immediately that if we let $T_n = \inf\{t : B_t \in [2^{-n}, 2^{-n+1}] \times [a_n, a_n]^{d-1}\}$ and pick a_n small enough, then $P_0(T_n < \infty) \leq 3^{-n}$. Now $\sum_{n=1}^{\infty} 3^{-n} = 3^{-1}(3/2) = 1/2$, so if we let $\tau = \inf\{t > 0 : B_t \notin G\}$ and $\sigma = \inf\{t > 0 : B_t \notin (-1, 1)^d\}$, then we have

$$P_0(\tau < \sigma) \leq \sum_{n=1}^{\infty} P_0(T_n < \infty) \leq \frac{1}{2}$$

Thus $P_0(\tau > 0) \geq P_0(\tau = \sigma) \geq 1/2$. Since $\tau = 0$ is measurable with respect to $\mathcal{F}_0^+$, we have $P_0(\tau = 0) = 1$. $\square$

A point $y \in \partial G$ is said to be a **regular point** if $P_y(\tau = 0) = 1$.

Theorem 9.5.7 *Let G be any open set. Suppose f is bounded and continuous and y is a regular point of ∂G . If $x_n \in G$ and $x_n \rightarrow y$, then $v(x_n) \rightarrow f(y)$.*

The first step is to show:

Lemma 9.5.8 *If $t > 0$, then $x \rightarrow P_x(\tau \leq t)$ is lower semicontinuous. That is, if $x_n \rightarrow x$, then*

$$\liminf_{n \rightarrow \infty} P_{x_n}(\tau \leq t) \geq P_x(\tau \leq t)$$

Proof By the Markov property

$$P_x(B_s \in G^c \text{ for some } s \in (\epsilon, t]) = \int p_\epsilon(x, y) P_y(\tau \leq t - \epsilon) dy$$

Since $y \rightarrow P_y(\tau \leq t - \epsilon)$ is bounded and measurable and

$$p_\epsilon(x, y) = (2\pi\epsilon)^{-d/2} e^{-|x-y|^2/2\epsilon}$$

it follows from the dominated convergence theorem that

$$x \rightarrow P_x(B_s \in G^c \text{ for some } s \in (\epsilon, t])$$

is continuous for each $\epsilon > 0$. Letting $\epsilon \downarrow 0$ shows that $x \rightarrow P_x(\tau \leq t)$ is an increasing limit of continuous functions and, hence, lower semicontinuous. $\square$

If y is regular for G and $t > 0$, then $P_y(\tau \leq t) = 1$, so it follows from Lemma 9.5.8 that if $x_n \rightarrow y$, then

$$\liminf_{n \rightarrow \infty} P_{x_n}(\tau \leq t) \geq 1 \quad \text{for all } t > 0$$

With this established, it is easy to complete the proof. Since f is bounded and continuous, it suffices to show that if $D(y, \delta) = \{x : |x - y| < \delta\}$, then

Lemma 9.5.9 *If y is regular for G and $x_n \rightarrow y$, then for all $\delta > 0$*

$$P_{x_n}(\tau < \infty, B_\tau \in D(y, \delta)) \rightarrow 1$$

Proof Let $\epsilon > 0$ and pick t so small that

$$P_0\left(\sup_{0 \leq s \leq t} |B_s| > \frac{\delta}{2}\right) < \epsilon$$

Since $P_{x_n}(\tau \leq t) \rightarrow 1$ as $x_n \rightarrow y$, it follows from the choices above that

$$\begin{aligned} \liminf_{n \rightarrow \infty} P_{x_n}(\tau < \infty, B_\tau \in D(y, \delta)) &\geq \liminf_{n \rightarrow \infty} P_{x_n}\left(\tau \leq t, \sup_{0 \leq s \leq t} |B_s - x_n| \leq \frac{\delta}{2}\right) \\ &\geq \liminf_{n \rightarrow \infty} P_{x_n}(\tau \leq t) - P_0\left(\sup_{0 \leq s \leq t} |B_t| > \frac{\delta}{2}\right) > 1 - \epsilon \end{aligned}$$

Since ϵ was arbitrary, this proves the lemma and hence Theorem 9.5.7. $\square$

The next result shows that if G^c is not too small near y , then y is regular.

Theorem 9.5.10 (Cone Condition) *If there is a cone V having vertex y and an $r > 0$ such that $V \cap D(y, r) \subset G^c$, then y is a regular point.*

Proof The first thing to do is to define a **cone with vertex y , pointing in direction v , with opening a** as follows:

$$V(y, v, a) = \{x : x = y + \theta(v + z), \text{ where } \theta \in (0, \infty), z \perp v, \text{ and } |z| < a\}$$

Now that we have defined a cone, the rest is easy. Since the normal distribution is spherically symmetric,

$$P_y(B_t \in V(y, v, a)) = \epsilon_a > 0$$

where ϵ_a is a constant that depends only on the opening a . Let $r > 0$ be such that $V(y, v, a) \cap D(y, r) \subset G^c$. The continuity of Brownian paths implies

$$\lim_{t \rightarrow 0} P_y \left(\sup_{s \leq t} |B_s - y| > r \right) = 0$$

Combining the last two results with a trivial inequality, we have

$$\epsilon_a \leq \liminf_{t \downarrow 0} P_y(B_t \in G^c) \leq \lim_{t \downarrow 0} P_y(\tau \leq t) \leq P_y(\tau = 0)$$

and it follows from Blumenthal's zero-one law that $P_y(\tau = 0) = 1$. □

9.5.1 Exit Distributions

Half-space. Let $\tau = \inf\{t : B_t \notin H\}$, where $H = \{(x, y) : x \in \mathbf{R}^{d-1}, y > 0\}$. For $\theta \in \mathbf{R}^{d-1}$ let

$$h_\theta(x, y) = \frac{C_d y}{(|x - \theta|^2 + y^2)^{d/2}} \quad (9.5.2)$$

where C_d is chosen so that $\int d\theta h_\theta(0, 1) = 1$.

Lemma 9.5.11 $\Delta h_\theta = 0$ in H

Proof (a) Ignoring C_d and differentiating gives

$$\begin{aligned} D_{x_i} h_\theta &= -\frac{d}{2} \cdot \frac{2(x_i - \theta_i)y}{(|x - \theta|^2 + y^2)^{(d+2)/2}} \\ D_{x_i x_i} h_\theta &= -d \frac{y}{(|x - \theta|^2 + y^2)^{(d+2)/2}} + \frac{d(d+2)(x_i - \theta_i)^2 y}{(|x - \theta|^2 + y^2)^{(d+4)/2}} \\ D_y h_\theta &= \frac{1}{(|x - \theta|^2 + y^2)^{d/2}} - d \cdot \frac{y^2}{(|x - \theta|^2 + y^2)^{(d+2)/2}} \\ D_{yy} h_\theta &= -d \cdot \frac{y + 2y}{(|x - \theta|^2 + y^2)^{(d+2)/2}} + \frac{d(d+2)y^3}{(|x - \theta|^2 + y^2)^{(d+4)/2}} \end{aligned}$$

Adding up, we see that

$$\sum_{i=1}^{d-1} D_{x_i x_i} h_\theta + D_{yy} h_\theta = \frac{\{(-d)(d-1) + (-d) \cdot 3\}y}{(|x - \theta|^2 + y^2)^{(d+2)/2}} + \frac{d(d+2)y(|x - \theta|^2 + y^2)}{(|x - \theta|^2 + y^2)^{(d+4)/2}} = 0$$

which completes the proof. $\square$

Theorem 9.5.12 *Let f be bounded and continuous and*

$$u(x, y) = \int d\theta h_\theta(x, y) f(\theta, 0)$$

u satisfies the Dirichlet problem in H with boundary value f . From this it follows that $h_\theta(x, y)$ gives the probability density $P_{x,y}(B(\tau_H) = (\theta, 0))$.

Proof Interchanging the derivative and integral, we see that $\Delta u = 0$. The next step is to show that it satisfies the boundary condition.

Clearly, $\int d\theta h_\theta(x, y)$ is independent of x . To show that it is independent of y , let $x = 0$ and change variables $\theta_i = y\phi_i$ for $1 \leq i \leq d-1$ to get

$$\int d\theta h_\theta(0, y) = \int \frac{C_d y}{(y^2|\phi|^2 + y^2)^{d/2}} \cdot y^{d-1} d\phi = \int d\phi h_\phi(0, 1) = 1$$

Changing variables $\theta_i = x_i + r_i y$ and using dominated convergence

$$\int_{D(x, \epsilon)^c} d\theta h_\theta(x, y) = \int_{D(0, \epsilon/y)^c} dr h_r(0, 1) \rightarrow 0$$

Since $P_{x,y}(\tau < \infty) = 1$ for all $x \in H$, the last conclusion follows from Theorem 9.5.2. $\square$

When $d = 2$

$$h_\theta(0, y) = \frac{y}{\pi(|\theta|^2 + y^2)}$$

is Cauchy density. To see why this should be true, start the Brownian motion at $(0, 0)$ and let $T_y = \inf\{t : B_t^2 = y\}$. The strong Markov property implies that $B^1(T_y)$ has independent increments. Brownian scaling implies $B^1(T_y) =_d y B^1(T_1)$. Since $B^1(T_y) =_d -B^1(T_y)$, the distribution must be the symmetric stable law with $\alpha = 1$.

Exit distributions for the ball. Let $D = \{x : |x| < 1\}$, let $\tau_D = \inf\{t : B_t \notin D\}$, and define the Poisson kernel by

$$k_y(x) = \frac{1 - |x|^2}{|x - y|^d} \quad (9.5.3)$$

Theorem 9.5.13 *If f is bounded and continuous, then*

$$E_x f(B_\tau) = u(x) = \int_{\partial D} k_y(x) f(y) \pi(dy) \quad (9.5.4)$$

where π is surface measure on ∂D normalized to be a probability measure.

The proof technique is the same as for the result in the half-space. We show that u satisfies the Dirichlet problem with boundary condition f . We leave it to the reader to show that for fixed y , $\Delta k_y(x) = 0$ and then complete the proof as previously.

Exercises

- 9.5.1 Suppose h is bounded and harmonic on $\mathbf{R}^d$. Prove that h is constant.
- 9.5.2 Suppose $f \in C^2$ is superharmonic, takes values in $[0, \infty]$ and is continuous. Show that in dimensions 1 and 2, f must be constant. Give an example of the form $f(x) = g(|x|)$ in $d \geq 3$.
- 9.5.3 Let B_t be a d -dimensional Brownian motion and let $\tau = \inf\{t : B_t \notin G\}$. Suppose G is a connected open set, all boundary points are regular, and $P_x(\tau < \infty) < 1$ in G . Show that all bounded solutions of the Dirichlet problem with boundary value 0 on ∂G have the form $u(x) = cP_x(\tau = \infty)$. Hint: Define $\bar{u} = u + \|u\|_\infty P_x(\tau = \infty)$ in G , $\bar{u} = 0$ on G^c , and show that $u(B_y)$ is a submartingale.
- 9.5.4 Let $\tau = \inf\{t : B_t \notin G\}$ and consider now the Dirichlet problem in a connected open set G in $d \geq 3$, where all boundary points are regular and which has $P_x(\tau < \infty) < 1$ in G . Suppose that the boundary condition $f \equiv 0$. Generalize the previous result to conclude that all bounded solutions have the form $u(x) = cP_x(\tau = \infty)$.
- 9.5.5 Define a flat cone $\bar{V}(y, v, a)$ to be the intersection of $V(y, v, a)$ with a $d-1$ dimensional hyperplane that contains the line $\{y + \theta v : \theta \in \mathbf{R}\}$. Show that Theorem 9.5.10 remains true if “cone” is replaced by “flat cone.”

9.6 Green's Functions and Potential Kernels

Let $D = B(0, r) = \{y : |y| < r\}$ the ball of radius r centered at 0. B_t will return to D i.o. in $d \leq 2$ but not in $d \geq 3$. In this section we will investigate the occupation time $\int_0^\infty 1_D(B_t) dt$. The first step is to show that for any x

$$P_x \left(\int_0^\infty 1_D(B_t) dt = \infty \right) = 1 \quad \text{in } d \leq 2, \quad (9.6.1)$$

$$E_x \int_0^\infty 1_D(B_t) dt < \infty \quad \text{in } d \geq 3. \quad (9.6.2)$$

Proof Let $T_0 = 0$ and $G = B(0, 2r)$. For $k \geq 1$, let

$$\begin{aligned} S_k &= \inf\{t > T_{k-1} : B_t \in D\} \\ T_k &= \inf\{t > S_k : B_t \in G\} \end{aligned}$$

Writing τ for T_1 and using the strong Markov property, we get for $k \geq 1$

$$P_x \left(\int_{S_k}^{T_k} 1_D(B_t) dt \geq s \middle| \mathcal{F}_{S_k} \right) = P_{B(S_k)} \left(\int_0^\tau 1_D(B_t) dt \geq s \right) = H(s)$$

From the strong Markov property it follows that

$$\int_{S_k}^{T_k} 1_D(B_t) dt \quad \text{are i.i.d.}$$

Since these random variables have positive mean, it follows from the strong law of large numbers that

$$\int_0^\infty 1_D(B_t) dt \geq \lim_{n \rightarrow \infty} \sum_{k=1}^n \int_{S_k}^{T_k} 1_D(B_t) dt = \infty \quad \text{a.s.}$$

proving (9.6.1).

If f is a nonnegative function, then Fubini's theorem implies

$$\begin{aligned} E_x \int_0^\infty f(B_t) dt &= \int_0^\infty E_x f(B_t) dt = \int_0^\infty \int p_t(x, y) f(y) dy dt \\ &= \int \int_0^\infty p_t(x, y) dt f(y) dy \end{aligned}$$

where $p_t(x, y) = (2\pi t)^{-d/2} e^{-|x-y|^2/2t}$ is the transition density for Brownian motion. As $t \rightarrow \infty$, $p_t(x, y) \sim (2\pi t)^{-d/2}$, so if $d \leq 2$, then $\int p_t(x, y) dt = \infty$. When $d \geq 3$, changing variables $t = |x - y|^2/2s$ gives

$$\begin{aligned} \int_0^\infty p_t(x, y) dt &= \int_0^\infty \frac{1}{(2\pi t)^{d/2}} e^{-|y-x|^2/2t} dt \\ &= \int_\infty^0 \left(\frac{s}{\pi |x-y|^2} \right)^{d/2} e^{-s} \left(-\frac{|x-y|^2}{2s^2} \right) ds \\ &= \frac{|x-y|^{2-d}}{2\pi^{d/2}} \int_0^\infty s^{(d/2)-2} e^{-s} ds \\ &= \frac{\Gamma(\frac{d}{2} - 1)}{2\pi^{d/2}} |x-y|^{2-d} \end{aligned} \tag{9.6.3}$$

where $\Gamma(\alpha) = \int_0^\infty s^{\alpha-1} e^{-s} ds$ is the usual gamma function. If we define

$$G(x, y) = \int_0^\infty p_t(x, y) dt$$

then in $d \geq 3$, $G(x, y) < \infty$ for $x \neq y$, and

$$(2.4) \quad E_x \int_0^\infty f(B_t) dt = \int G(x, y) f(y) dy$$

To complete the proof now, we observe that taking $f = 1_D$ with $D = B(0, r)$ and changing to polar coordinates

$$\int_D G(0, y) dy = \int_0^r s^{d-1} C_d s^{2-d} ds = \frac{C_d}{2} r^2 < \infty$$

To extend the last conclusion to $x \neq 0$, observe that applying the strong Markov property at the exit time τ from $B(0, |x|)$ for a Brownian motion starting at 0 and using the rotational symmetry we have

$$E_x \int_0^\infty 1_D(B_s) ds = E_0 \int_\tau^\infty 1_D(B_s) ds \leq E_0 \int_0^\infty 1_D(B_s) ds$$

which completes the proof. $\square$

We call $G(x, y)$ the **potential kernel**, because $G(\cdot, y)$ is the electrostatic potential of a unit charge at y . See chapter 3 of Port and Stone (1978) for more on this. In $d \leq 2$, $\int_0^\infty p_t(x, y) dt \equiv \infty$, so we have to take another approach to define a useful G :

$$G(x, y) = \int_0^\infty (p_t(x, y) - a_t) dt \quad (9.6.4)$$

where the a_t are constants we will choose to make the integral converge (at least when $x \neq y$). To see why this modified definition might be useful note that if $\int f(y) dy = 0$, then

$$\int G(x, y) f(y) dy = \lim_{t \rightarrow \infty} \int_0^t E_x f(B_s) ds$$

When $d = 1$ we let $a_t = p_t(0, 0)$. With this choice,

$$G(x, y) = \frac{1}{\sqrt{2\pi}} \int_0^\infty (e^{-(y-x)^2/2t} - 1) t^{-1/2} dt$$

and the integral converges, since the integrand is ≤ 0 and $\sim -(y-x)^2/2t^{3/2}$ as $t \rightarrow \infty$. Changing variables $t = (y-x)^2/2u$ gives

$$\begin{aligned} G(x, y) &= \frac{1}{\sqrt{2\pi}} \int_0^\infty (e^{-u} - 1) \left(\frac{2u}{(y-x)^2} \right)^{1/2} \frac{-(y-x)^2}{2u^2} du \\ &= -\frac{|y-x|}{2\sqrt{\pi}} \int_0^\infty \left(\int_0^u e^{-s} ds \right) u^{-3/2} du \\ &= -\frac{|y-x|}{\sqrt{\pi}} \int_0^\infty ds e^{-s} \int_s^\infty \frac{u^{-3/2}}{2} du \\ &= -\frac{|y-x|}{\sqrt{\pi}} \int_0^\infty ds e^{-s} s^{-1/2} = -|y-x| \end{aligned} \quad (9.6.5)$$

since

$$\int_0^\infty ds e^{-s} s^{-1/2} = \int_0^\infty dr e^{-r^2/2} \sqrt{2} = \frac{1}{2} \sqrt{2} \cdot \sqrt{2\pi} = \sqrt{\pi}$$

The computation is almost the same for $d = 2$. The only thing that changes is the choice of a_t . If we try $a_t = p_t(0, 0)$ again, then for $x \neq y$ the integrand $\sim -t^{-1}$ as $t \rightarrow 0$ and the integral diverges, so we let $a_t = p_t(0, e_1)$, where $e_1 = (1, 0)$. With this choice of a_t , we get

$$\begin{aligned}
G(x, y) &= \frac{1}{2\pi} \int_0^\infty (e^{-|x-y|^2/2t} - e^{-1/2t}) t^{-1} dt \\
&= \frac{1}{2\pi} \int_0^\infty \left(\int_{|x-y|^2/2t}^{1/2t} e^{-s} ds \right) t^{-1} dt \\
&= \frac{1}{2\pi} \int_0^\infty ds e^{-s} \int_{|x-y|^2/2s}^{1/2s} t^{-1} dt \\
&= \frac{1}{2\pi} \left(\int_0^\infty ds e^{-s} \right) (-\log(|x-y|^2)) \\
&= \frac{-1}{\pi} \log(|x-y|)
\end{aligned} \tag{9.6.6}$$

To sum up, the potential kernels are given by

$$G(x, y) = \begin{cases} (\Gamma(d/2 - 1)/2\pi^{d/2}) \cdot |x - y|^{2-d} & d \geq 3 \\ (-1/\pi) \cdot \log(|x - y|) & d = 2 \\ -1 \cdot |x - y| & d = 1 \end{cases} \tag{9.6.7}$$

The reader should note that in each case $G(x, y) = C\varphi(|x - y|)$, where φ is the harmonic function we used in Section 9.1. This is, of course, no accident. $x \rightarrow G(x, 0)$ is obviously spherically symmetric and, as we will see, satisfies $\Delta G(x, 0) = 0$ for $x \neq 0$, so the results above imply $G(x, 0) = A + B\varphi(|x|)$.

The formulas above correspond to $A = 0$, which is nice and simple. But what about the weird-looking B 's? What is special about them? The answer is simple: They are chosen to make $\frac{1}{2}\Delta G(x, 0) = -\delta_0$ (a point mass at 0) in the distributional sense. It is easy to see that this happens in $d = 1$. In that case, $\phi(x) = |x|$ then

$$\phi'(x) = \begin{cases} 1 & x > 0 \\ -1 & x < 0 \end{cases}$$

so if $B = -1$, then $B\phi''(x) = -2\delta_0$. More sophisticated readers can check that this is also true in $d \geq 2$.

9.7 Poisson's Equation

In this section, we study the following equation in a bounded open set G :

- (a) $u \in C^2$ and $\frac{1}{2}\Delta u = -g$ in G , where g is bounded and continuous.
- (b) At each point of ∂G , u is continuous and $u = 0$.

We can add a solution of the Dirichlet problem to replace $u = 0$ in (b) by $u = f$. It follows from Theorem 9.1.1 that

Theorem 9.7.1 *Let $\tau = \inf\{t > 0 : B_t \notin G\}$. If u satisfies (a), then*

$$M_t = u(B_t) + \int_0^t g(B_s) ds$$

is a local martingale on $[0, \tau)$.

The next step is to prove a uniqueness result.

Theorem 9.7.2 *If there is a solution of (a) that is bounded, it must be*

$$v(x) \equiv E_x \left(\int_0^\tau g(B_t) dt \right)$$

Note that if $g \equiv 1$, the solution is $v(x) = E_x \tau$.

Proof If u satisfies (a), then M_t defined is a local martingale on $[0, \tau)$. If G is bounded, then $E_x \tau < \infty$ for all $x \in G$. If u and g are bounded, then for $t < \tau$

$$|M_t| \leq \|u\|_\infty + \tau \|g\|_\infty$$

Since M_t is dominated by an integrable random variable, it follows that

$$M_\tau \equiv \lim_{t \uparrow \tau} M_t = \int_0^\tau g(B_t) dt$$

and $u(x) = E_x(M_\tau)$. □

As in our treatment of the Dirichlet problem:

Theorem 9.7.3 *If $v \in C^2$, then it satisfies (a).*

Proof Let $x \in D$ and $B(x, r) \subset D$. The strong Markov property implies

$$v(x) = E_x \int_0^{\tau_B} g(B_s) ds + E_x v(B(\tau_B))$$

Using Taylor's theorem

$$\begin{aligned} E_x v(B(\tau_B)) - v(x) &= E_x \sum_{i=1}^d \frac{\partial v}{\partial x_i}(x) [B_i(\tau_B) - x_i] \\ &\quad + \frac{1}{2} E_x \sum_{1 \leq i, j \leq d} \frac{\partial^2 v}{\partial x_i \partial x_j}(x) [B_i(\tau_B) - x_i] [B_j(\tau_B) - x_j] + o(r^2) \end{aligned}$$

Since for $i \neq j$, $B_i(t)$, $B_i^2(t) - t$ and $B_i(t)B_j(t)$ are martingales, we have

$$E_x v(B(\tau_B)) - v(x) = \frac{1}{2} \Delta v(x) E_x \tau_B + o(r^2)$$

On the other hand, $E_x \int_0^{\tau_B} g(B_s) ds = [g(x) + o(1)] E_x \tau_B$, so letting $r \rightarrow 0$ we have $(1/2) \Delta v = -g$ □

After the extensive discussion in the last section on the Dirichlet problem, the conditions needed to guarantee that the boundary conditions hold should come as no surprise.

Theorem 9.7.4 *Suppose that G and g are bounded. Let y be a regular point of ∂G . If $x_n \in G$ and $x_n \rightarrow y$, then $v(x_n) \rightarrow 0$.*

Proof We begin by observing:

(i) It follows from Lemma 9.5.9 that if $\epsilon > 0$, then $P_{x_n}(\tau > \epsilon) \rightarrow 0$.

(ii) If G is bounded, then $C = \sup_x E_x \tau < \infty$ and, hence, $\|v\|_\infty \leq C \|g\|_\infty < \infty$.

Let $\epsilon > 0$. Beginning with some elementary inequalities then using the Markov property, we have

$$\begin{aligned} |v(x_n)| &\leq E_{x_n} \left(\int_0^{\tau \wedge \epsilon} |g(B_s)| ds \right) + E_{x_n} \left(\left| \int_\epsilon^\tau g(B_s) ds \right| ; \tau > \epsilon \right) \\ &\leq \epsilon \|g\|_\infty + E_{x_n} (|v(B_\epsilon)|; \tau > \epsilon) \leq \epsilon \|g\|_\infty + \|v\|_\infty P_{x_n}(\tau > \epsilon) \end{aligned}$$

Letting $n \rightarrow \infty$, and using (i) and (ii) proves the desired result, since ϵ is arbitrary. $\square$

Last, but not least, we come to the question of proving $v \in C^2$. For these developments we will assume that g is defined on $\mathbf{R}^d$ not just in G and has compact support. Recall we are supposing G is bounded and notice that the values of g on G^c are irrelevant, so there will be no loss of generality if we later want to suppose that $\int g(x) dx = 0$. We begin with the case $d \geq 3$, because in this case implies

$$\bar{w}(x) = E_x \int_0^\infty |g(B_t)| dt < \infty$$

and, moreover, is a bounded function of x . This means we can define

$$w(x) = E_x \int_0^\infty g(B_t) dt,$$

use the strong Markov property to conclude

$$w(x) = E_x \int_0^\tau g(B_t) dt + E_x w(B_\tau),$$

and change notation to get

$$v(x) = w(x) - E_x w(B_\tau). \quad (9.7.1)$$

Theorem 9.5.4 tells us that the second term is C^∞ in G , so to verify that $v \in C^2$ we need only prove that w is a task in $d \geq 3$ that is made simpler by the fact that gives the following explicit formula

$$w(x) = C_d \int |x - y|^{2-d} g(y) dy$$

The details are tedious, so we will content ourselves to state the result.

Theorem 9.7.5 *In $d \geq 2$ if g is Hölder continuous, then w is C^2 . In $d = 1$ it is sufficient to suppose g is continuous.*

The last result settles the question of smoothness in $d \geq 3$. To extend the result to $d \leq 2$, we use the potential kernels defined in (9.6.4). Using this interpretation of w , it is easy to see that (9.7.1) holds, so again our problem is reduced to proving that w is C^2 , which is a problem in calculus. Once all the computations are done, we find that Theorem 9.7.5 holds in $d = 2$ and that in $d = 1$, it is sufficient to assume that g is continuous. The reader can find proof either in Port and Stone (1978), pp. 116-117, or in Gilbarg and Trudinger (1977), pp. 53-55.

9.7.1 Occupation Times

Half space. To explain why we want to define the potential kernels, we return to Brownian motion in the half space, $H = \{y : y_d > 0\}$. Let $\tau = \inf\{t : B_t \notin H\}$, and let

$$\bar{y} = (y_1, \dots, y_{d-1}, -y_d)$$

be the reflection of y through the plane $\{y \in \mathbf{R}^d : y_d = 0\}$.

Theorem 9.7.6 *If $x \in H$, $f \geq 0$ has compact support, and $\{x : f(x) > 0\} \subset H$, then*

$$E_x \left(\int_0^\tau f(B_t) dt \right) = \int G(x, y) f(y) dy - \int G(x, \bar{y}) f(y) dy$$

Proof Using Fubini's theorem, which is justified since $f \geq 0$, then using the reflection principle and the fact that $\{x : f(x) > 0\} \subset H$, we have

$$\begin{aligned} E_x \int_0^\tau f(B_t) dt &= \int_0^\infty E_x(f(B_t); \tau > t) dt \\ &= \int_0^\infty \int_H (p_t(x, y) - p_t(x, \bar{y})) f(y) dy dt \\ &= \int_0^\infty \int_H (p_t(x, y) - a_t) f(y) dy dt \\ &\quad - \int_0^\infty \int_H (p_t(x, \bar{y}) - a_t) f(y) dy dt \\ &= \int G(x, y) f(y) dy - \int G(x, \bar{y}) f(y) dy \end{aligned}$$

The compact support of f and the formulas for G imply that $\int |G(x, y) f(y)| dy$ and $\int |G(x, \bar{y}) f(y)| dy$ are finite, so the last two equalities are valid. $\square$

The proof given here simplifies considerably in the case $d \geq 3$; however, part of the point of this proof is that, with the definition we have chosen for G in the recurrent case, the formulas and proofs can be the same for all d .

Let $G_H(x, y) = G(x, y) - G(x, \bar{y})$. We think of $G_H(x, y)$ as the “expected occupation time (density) at y for a Brownian motion starting at x and killed when it leaves H .” The rationale for this interpretation is that (for suitable f)

$$E_x \left(\int_0^\tau f(B_t) dt \right) = \int G_H(x, y) f(y) dy$$

The answer is very simple in $d = 1$.

$$G_H(x, y) = 2(x \wedge y) \quad \text{for all } x, y > 0 \quad (9.7.2)$$

It is somewhat surprising that $y \rightarrow G_H(x, y)$ is constant $= 2x$ for $y \geq x$, that is, all points $y > x$ have the same expected occupation time!

Proof $G(x, y) = -|x - y|$, so $G_H(x, y) = -|x - y| + |x + y|$. Separating things into cases, we see that

$$G_H(x, y) = \begin{cases} -(x - y) + (x + y) = 2y & \text{when } 0 < y < x \\ -(y - x) + (x + y) = 2x & \text{when } x < y \end{cases}$$

which gives the desired formula. $\square$

Ball. Let $G(x, y)$ be the Green's functions defined in (9.6.7).

Theorem 9.7.7 *If g is bounded and measurable, then*

$$E_x \int_0^\tau g(B_t) dt = \int G_D(x, y) g(y) dy$$

where

$$G_D(x, y) = G(x, y) - \int \frac{1 - |x|^2}{|x - z|^d} G(z, y) \pi(dz) \quad (9.7.3)$$

This follows from (9.7.1) and (9.5.4). This formula can be simplified considerably. Turning to page 123 in Folland (1976), we find.

Theorem 9.7.8 *In $d \geq 3$, if $0 < |y| < 1$, then*

$$G_D(x, y) = G(x, y) - |y|^{2-d} G(x, y/|y|^2)$$

Proof By (9.7.1) it is enough to show that (i) the second term is harmonic in D and (ii) $= G(x, y)$ on the boundary. $y \rightarrow y/|y|^2$ maps points $y \in D$ to D^c , so (i) holds. To prove (ii), we note

$$\begin{aligned} G(x, y) - |y|^{2-d} G(x, y/|y|^2) &= \frac{C_d}{|x - y|^{d-2}} - \frac{C_d}{|y|^{d-2}} \left| x - \frac{y}{|y|^2} \right|^{-(d-2)} \\ &= \frac{C_d}{|x - y|^{d-2}} - \frac{C_d}{|x|y| - y|y|^{-1}|^{d-2}} = 0 \end{aligned}$$

The last equality follows from a fact useful for the next proof:

Lemma 9.7.9 *If $|x| = 1$, then $|x|y| - y|y|^{-1}| = |x - y|$.*

Using $|z|^2 = z \cdot z$ and then $|x|^2 = 1$, we have

$$\begin{aligned} ||x|y| - y|y|^{-1}|^2 &= |x|^2|y|^2 - 2x \cdot y + 1 \\ &= |y|^2 - 2x \cdot y + |x|^2 = |x - y|^2 \end{aligned}$$

which completes the proof. □

Again turning to page 123 of Folland (1976) we “guess”

Theorem 9.7.10 *In $d = 2$ if $0 < |y| < 1$, then*

$$G_D(x, y) = \frac{-1}{\pi} \left(\ln |x - y| - \ln |x|y| - y|y|^{-1}| \right)$$

We leave the proof to the reader.

In $d = 1$, where $D = (-1, 1)$, (9.7.3) tells us that

$$\begin{aligned} G_D(x, y) &= G(x, y) - \frac{x+1}{2} G(1, y) - \frac{1-x}{2} G(-1, y) \\ &= -|x - y| + \frac{x+1}{2} (1 - y) + \frac{1-x}{2} (y + 1) \\ &= -|x - y| + 1 - xy \end{aligned}$$

Considering the two cases $x \geq y$ and $x \leq y$ leads to

$$G_D(x, y) = \begin{cases} (1-x)(1+y) & -1 \leq y \leq x \leq 1 \\ (1-y)(1+x) & -1 \leq x \leq y \leq 1 \end{cases} \quad (9.7.4)$$

Geometrically, if we fix y , then $x \rightarrow G_D(x, y)$ is determined by the conditions that it is linear on $[0, y]$ and on $[y, 1]$ with $G_D(0, y) = G_D(1, y) = 0$, and $(1/2)G_D''(y, y) = -\delta_y$, a point mass at y . That is, G_D' is discontinuous at y and $G_D'(y-, y) - G_D'(y+, y) = 2$.

Exercise 9.7.1 Let G be a bounded open set and let $\tau = \inf\{t : B_t \notin G\}$. What equation should we solve to find $w(x) = E_x \tau^2$?

9.8 Schrödinger Equation

The developments here follow Chung and Zhao (1995). In this section we will consider what happens when we add cu where c is a bounded continuous function to the left-hand side of the Dirichlet problem. That is, given a bounded continuous function f on ∂G , we will study

(a) $u \in C^2$ and $\frac{1}{2}\Delta u + cu = 0$ in G .

(b) At each point of ∂G , u is continuous and $u = f$.

As always, the first step in solving the PDE is to find a local martingale.

Theorem 9.8.1 Let $\tau = \inf\{t > 0 : B_t \notin G\}$. If u satisfies (a), then

$$M_t = u(B_t) \exp\left(\int_0^t c(B_s) ds\right)$$

is a local martingale on $[0, \tau)$.

Proof Let $c_t = \int_0^t c(B_s) ds$. Applying the version of Itô's from (9.4.1) gives

$$\begin{aligned} u(B_t) \exp(c_t) - u(B_0) &= \int_0^t \exp(c_s) \nabla u(B_s) \cdot dB_s + \int_0^t u(B_s) \exp(c_s) c(B_s) ds \\ &\quad + \frac{1}{2} \int_0^t \Delta u(B_s) \exp(c_s) ds \end{aligned}$$

for $t < \tau$. This proves the result, since $\frac{1}{2}\Delta u + cu = 0$, and using theorem 7.6.4 the first term on the right-hand side is a local martingale on $[0, \tau)$. $\square$

At this point, the reader might expect that the next step, as it has been five times before, is to assume that everything is bounded and conclude that if there is a solution that is bounded, it must be

$$v(x) \equiv E_x(f(B_\tau) \exp(c_\tau))$$

We will not do this, however, because the following simple example shows that this result is false.

Example 9.8.2 Let $d = 1$, $G = (-a, a)$, $c \equiv \gamma$, and $f \equiv 1$. The equation we are considering is

$$\frac{1}{2}u'' + \gamma u = 0 \quad u(a) = u(-a) = 1$$

The general solution is $A \cos bx + B \sin bx$, where $b = \sqrt{2\gamma}$. So if we want the boundary condition to be satisfied, we must have

$$1 = A \cos ba + B \sin ba$$

$$1 = A \cos(-ba) + B \sin(-ba) = A \cos ba - B \sin ba$$

Adding the two equations and then subtracting them it follows that

$$2 = 2A \cos ba \quad 0 = 2B \sin ba$$

From this we see that $B = 0$ always works and we may or may not be able to solve for A .

If $\cos ba = 0$, then there is no solution. If $\cos ba \neq 0$, then $u(x) = \cos bx / \cos ba$ is a solution. We will see later that if $ab < \pi/2$, then

$$v(x) = \cos bx / \cos ba$$

However, this cannot possibly hold for $ab > \pi/2$ since $v(x) \geq 0$, while the right-hand side is < 0 for some values of x .

We will see that the trouble with the last example is that if $ab > \pi/2$, then $c \equiv \gamma$ is too large, or to be precise, if we let

$$w(x) = E_x \exp \left(\int_0^\tau c(B_s) ds \right)$$

then $w \equiv \infty$ in $(-a, a)$. The rest of this section is devoted to showing that if $w \not\equiv \infty$, then “everything is fine.” The development will require several stages. The first step is to show:

Lemma 9.8.3 Let $\theta > 0$. There is a $\mu > 0$ so that if H is an open set with Lebesgue measure $|H| \leq \mu$ and $\tau_H = \inf\{t > 0 : B_t \notin H\}$, then

$$\sup_x E_x (\exp(\theta \tau_H)) \leq 2$$

Proof Pick $\gamma > 0$ so that $e^{\theta\gamma} \leq 4/3$. Clearly,

$$P_x(\tau_H > \gamma) \leq \int_H \frac{1}{(2\pi\gamma)^{d/2}} e^{-|x-y|^2/2\gamma} dy \leq \frac{|H|}{(2\pi\gamma)^{d/2}} \leq \frac{1}{4}$$

if we pick μ small enough. Using the Markov property, we can conclude that

$$\begin{aligned} P_x(\tau_H > k\gamma) &= E_x(P_{B_\gamma}(\tau_H > (k-1)\gamma); \tau_H > \gamma) \\ &\leq \frac{1}{4} \sup_y P_y(\tau_H > (k-1)\gamma) \end{aligned}$$

So it follows by induction that for all integers $k \geq 0$ we have

$$\sup_x P_x(\tau_H > k\gamma) \leq \frac{1}{4^k}$$

Since e^x is increasing, and $e^{\theta\gamma} \leq 4/3$ by assumption, we have

$$\begin{aligned} E_x \exp(\theta\tau_H) &\leq \sum_{k=1}^{\infty} \exp(\theta\gamma k) P_x((k-1)\gamma < \tau_H \leq k\gamma) \\ &\leq \sum_{k=1}^{\infty} \left(\frac{4}{3}\right)^k \frac{1}{4^{k-1}} = \frac{4}{3} \sum_{k=1}^{\infty} \frac{1}{3^{k-1}} = \frac{4}{3} \cdot \frac{1}{1 - \frac{1}{3}} = 2 \end{aligned}$$

Careful readers may have noticed that we left $\tau_H = 0$ out of the expected value. However, by the Blumenthal 0-1 law either $P_x(\tau_H = 0) = 0$, in which case our computation is valid, or $P_x(\tau_H = 0) = 1$, in which case $E_x \exp(\theta\tau_H) = 1$. $\square$

Let $c^* = \sup_x |c(x)|$. By Lemma 9.8.3, we can pick r_0 so small that if $T_r = \inf\{t : |B_t - B_0| > r\}$ and $r \leq r_0$, then $E_x \exp(c^*T_r) \leq 2$ for all x .

Lemma 9.8.4 *Let $2\delta \leq r_0$. If $D(x, 2\delta) \subset G$ and $y \in D(x, \delta)$, then*

$$w(y) \leq 2^{d+2} w(x)$$

The reason for our interest in this is that it shows $w(x) < \infty$ implies $w(y) < \infty$ for $y \in D(x, \delta)$.

Proof If $D(y, r) \subset G$, and $r \leq r_0$, then the strong Markov property implies

$$\begin{aligned} w(y) &= E_y[\exp(c_{T_r})w(B(T_r))] \leq E_y[\exp(c^*T_r)w(B(T_r))] \\ &= E_y[\exp(c^*T_r)] \int_{\partial D(y, r)} w(z) \pi(dz) \leq 2 \int_{\partial D(y, r)} w(z) \pi(dz) \end{aligned} \quad (9.8.1)$$

where π is surface measure on $\partial D(y, r)$ normalized to be a probability measure, since the exit time, T_r , and exit location, $B(T_r)$, are independent.

If $\delta \leq r_0$ and $D(y, \delta) \subset G$, multiplying the last inequality by r^{d-1} and integrating from 0 to δ gives

$$\frac{\delta^d}{d} w(y) \leq 2 \cdot \frac{1}{\sigma_d} \int_{D(y, \delta)} w(z) dz$$

where σ_d is the surface area of $\{x : |x| = 1\}$. Rearranging we have

$$\int_{D(y, \delta)} w(z) dz \geq 2^{-1} \frac{\delta^d}{C_o} w(y) \quad (9.8.2)$$

where $C_o = d/\sigma_d$ is a constant that depends only on d .

Repeating the argument in (9.8.1) with $y = x$ and using the fact that $c_{T_r} \geq -c^*T_r$ gives

$$\begin{aligned} w(x) &= E_x[\exp(c_{T_r})w(B(T_r))] \geq E_x[\exp(-c^*T_r)w(B(T_r))] \\ &= E_x[\exp(-c^*T_r)] \int_{\partial D(x, r)} w(z) \pi(dz) \end{aligned}$$

Since $1/x$ is convex, Jensen's inequality implies

$$E_x[\exp(-c^*T_r)] \geq 1/E_x[\exp(c^*T_r)] \geq 1/2$$

Combining the last two displays, multiplying by r^{d-1} , and integrating from 0 to 2δ , we get the lower bound

$$\frac{(2\delta)^d}{d} w(x) \geq 2^{-1} \cdot \frac{1}{\sigma_d} \int_{D(x, 2\delta)} w(z) dz$$

Rearranging and using $D(x, 2\delta) \supset D(y, \delta)$, $w \geq 0$, we have

$$w(x) \geq 2^{-1} \frac{C_o}{(2\delta)^d} \int_{D(y, \delta)} w(z) dz \quad (9.8.3)$$

where again $C_o = d/\sigma_d$. Combining (9.8.3) and (9.8.2), it follows that

$$\begin{aligned} &\geq 2^{-1} \frac{C_o}{(2\delta)^d} \int_{D(y, \delta)} w(z) dz \\ &\geq 2^{-1} \frac{C_o}{(2\delta)^d} \cdot 2^{-1} \frac{\delta^d}{C_o} w(y) = 2^{-(d+2)} w(y) \end{aligned}$$

which gives the desired result. $\square$

Lemma 9.8.4 and a simple covering argument lead to:

Theorem 9.8.5 *Let G be a connected open set. If $w \not\equiv \infty$, then*

$$w(x) < \infty \quad \text{for all } x \in G$$

Proof From Lemma 9.8.4, we see that if $w(x) < \infty$, $2\delta \leq r_0$, and $D(x, 2\delta) \subset G$, then $w < \infty$ on $D(x, \delta)$. From this result, it follows that $G_0 = \{x : w(x) < \infty\}$ is an open subset of G . To argue now that G_0 is also closed (when considered as a subset of G), we observe that if $2\delta < r_0$, $D(y, 3\delta) \subset G$, and we have $x_n \in G_0$ with $x_n \rightarrow y \in G$, then for n sufficiently large, $y \in D(x_n, \delta)$ and $D(x_n, 2\delta) \subset G$, so $w(y) < \infty$. $\square$

Before we proceed to the uniqueness result, we want to strengthen the last conclusion.

Theorem 9.8.6 *Let G be a connected open set with finite Lebesgue measure, $|G| < \infty$. If $w \not\equiv \infty$, then*

$$\sup_x w(x) < \infty$$

Proof Let $K \subset G$ be compact, so that $|G - K| < \mu$ the constant in Lemma 9.8.3 for $\theta = c^*$. For each $x \in K$ we can pick a δ_x so that $2\delta_x \leq r_0$ and $D(x, 2\delta_x) \subset G$. The open sets $D(x, \delta_x)$ cover K , so there is a finite subcover $D(x_i, \delta_{x_i})$, $1 \leq i \leq I$. Clearly,

$$\sup_{1 \leq i \leq I} w(x_i) < \infty$$

Lemma 9.8.4 implies $w(y) \leq 2^{d+2} w(x_i)$ for $y \in D(x_i, \delta_{x_i})$, so

$$M = \sup_{y \in K} w(y) < \infty$$

If $y \in H = G - K$, then $E_y(\exp(c^* \tau_H)) \leq 2$ by (6.3a), so using the strong Markov property

$$\begin{aligned} w(y) &= E_y(\exp(c_{\tau_H}); \tau_H = \tau) + E_y(\exp(c_{\tau_H}) w(B_{\tau_H}); B_{\tau_H} \in K) \\ &\leq 2 + M E_y(\exp(c_{\tau_H}); B_{\tau_H} \in K) \leq 2 + 2M \end{aligned}$$

we have the desired result. $\square$

With Theorem 9.8.6 established, we are now more than ready to prove our uniqueness result. To simplify the statements of the results that follow, we will now list the assumptions that we will make for the rest of the section.

(A1) G is a bounded connected open set.

(A2) f and c are bounded and continuous.

(A3) $w \not\equiv \infty$.

Theorem 9.8.7 *If there is a solution that is bounded, it must be*

$$v(x) \equiv E_x(f(B_\tau) \exp(c_\tau))$$

Proof Theorem 9.8.1 implies that $M_s = u(B_s) \exp(c_s)$ is a local martingale on $[0, \tau)$. Since f , c , and u are bounded, letting $s \uparrow \tau \wedge t$ and using the bounded convergence theorem gives

$$u(x) = E_x(f(B_\tau) \exp(c_\tau); \tau \leq t) + E_x(u(B_t) \exp(c_t); \tau > t)$$

Since f is bounded and $w(x) = E_x \exp(c_\tau) < \infty$, the dominated convergence theorem implies that as $t \rightarrow \infty$, the first term converges to $E_x(f(B_\tau) \exp(c_\tau))$. To show that the second term $\rightarrow 0$, we begin with the observation that since $\{\tau > t\} \in \mathcal{F}_t$, the definition of conditional expectation and the Markov property imply

$$\begin{aligned} E_x(u(B_t) \exp(c_t); \tau > t) &= E_x(E_x(u(B_t) \exp(c_t) | \mathcal{F}_t); \tau > t) \\ &= E_x(u(B_t) \exp(c_t) w(B_t); \tau > t) \end{aligned}$$

Now we claim that for all $y \in G$

$$w(y) \geq \exp(-c^*) P_y(\tau \leq 1) \geq \epsilon > 0$$

The first inequality is trivial. The last two follow easily from (A1).

Replacing $w(B_t)$ by ϵ ,

$$\begin{aligned} E_x(|u(B_t)| \exp(c_t); \tau > t) &\leq \epsilon^{-1} E_x(|u(B_t)| \exp(c_t); \tau > t) \\ &\leq \epsilon^{-1} \|u\|_\infty E_x(\exp(c_t); \tau > t) \rightarrow 0 \end{aligned}$$

as $t \rightarrow \infty$, by the dominated convergence theorem, since $w(x) = E_x \exp(c_\tau) < \infty$ and $P_x(\tau < \infty) = 1$. Going back to the first equation in the proof, we have shown $u(x) = v(x)$ and the proof is complete. $\square$

This completes our consideration of uniqueness. The next stage in our program, fortunately, is as easy as it always has been. Recall that here and in what follows we are assuming (A1)–(A3).

Theorem 9.8.8 *If $v \in C^2$, then it satisfies (a) in G .*

Proof Let $x \in D$ and $B(x, r) \subset D$. Let σ be the exit time from $B(x, r)$. The strong Markov property implies

$$v(x) = E_x v(B_\sigma \exp(c_\sigma))$$

Using Taylor's theorem

$$\begin{aligned} E_x v(B(\sigma) \exp(c_\sigma)) - v(x) &= E_x \sum_{i=1}^d \frac{\partial v}{\partial x_i}(x) [B_i(\tau_D) - x_i] \\ &\quad + \frac{1}{2} E_x \sum_{1 \leq i, j \leq d} \frac{\partial^2 v}{\partial x_i \partial x_j}(x) [B_i(\tau_D) - x_i] [B_j(\tau_D) - x_j] + o(r^2) \end{aligned}$$

Since for $i \neq j$, $B_i(t)$, $B_i^2(t) - t$ and $B_i(t)B_j(t)$ are martingales, we have

$$E_x v(B(\tau_D)) - v(x) = \frac{1}{2} \Delta v(x) E_x \tau_B + o(r^2)$$

On the other hand, $E_x \int_0^{\tau_B} g(B_s) ds = [g(x) + o(1)] E_x \tau_B$, so letting $r \rightarrow 0$ we have $(1/2)\Delta v = -g$. $\square$

Having proved Theorem 9.8.8, the next step is to consider the boundary condition. As in the last two sections, we need the boundary to be regular.

Theorem 9.8.9 *v satisfies (b) at each regular point of ∂G .*

Proof Let y be a regular point of ∂G . We showed in (4.5a) and (4.5b) that if $x_n \rightarrow y$, then $P_{x_n}(\tau \leq \delta) \rightarrow 1$ and $P_{x_n}(B_\tau \in D(y, \delta)) \rightarrow 1$ for all $\delta > 0$. Since c is bounded and f is bounded and continuous, the bounded convergence theorem implies that

$$E_{x_n}(\exp(c_\tau) f(B_\tau); \tau \leq 1) \rightarrow f(y)$$

To control the contribution from the rest of the space, we observe that if $|c| \leq M$, then using the Markov property and the boundedness of w established in Theorem 9.8.6, we have

$$\begin{aligned} E_{x_n}(\exp(c_\tau) f(B_\tau); \tau > 1) &\leq e^M \|f\|_\infty E_{x_n}(w(B_1); \tau > 1) \\ &\leq e^M \|f\|_\infty \|w\|_\infty P_{x_n}(\tau > 1) \rightarrow 0 \end{aligned}$$

which completes the proof. $\square$

This brings us finally to the problem of determining when v is smooth enough to be a solution.

Theorem 9.8.10 *If in addition to (A1)–(A3), c is Hölder continuous, then $v \in C^2$ and, hence, satisfies (a).*

Return to Example 9.8.2. Recall $d = 1$, $G = (-a, a)$, $c(x) \equiv \gamma > 0$, and $f \equiv 1$.

Theorem 9.8.11 *Let $\tau = \inf\{t : B_t \notin (-a, a)\}$. If $0 < \gamma < \pi^2/8a^2$, then*

$$E_x e^{\gamma \tau} = \frac{\cos(x\sqrt{2\gamma})}{\cos(a\sqrt{2\gamma})}$$

If $\gamma \geq \pi^2/8a^2$, then $E_x e^{\gamma \tau} \equiv \infty$.

Proof By results in Example 9.8.2

$$u(x) = \cos(x\sqrt{2\gamma}) / \cos(a\sqrt{2\gamma})$$

is a nonnegative solution of

$$\frac{1}{2}u'' + \gamma u = 0 \quad u(-a) = u(a) = 1$$

To check that $w \not\equiv \infty$, we observe that Theorem 9.8.1 implies that $M_t = u(B_t)e^{\gamma t}$ is a local martingale on $[0, \tau)$. If we let T_n be a sequence of stopping times that reduces M_t , then

$$u(x) = E_x \left(u(B_{T_n \wedge n}) e^{\gamma(T_n \wedge n)} \right)$$

Letting $n \rightarrow \infty$ and noting $T_n \wedge n \uparrow \tau$, it follows from Fatou's lemma that

$$u(x) \geq E_x e^{\gamma \tau}$$

The last equation implies $w \not\equiv \infty$, so Theorem 9.8.8 implies the result for $\gamma < \pi^2/8a^2$.

To see that $w(x) \equiv \infty$ when $\gamma \geq \pi^2/8a^2$, suppose not. In this case, (6.3) implies $v(x) = E_x(f(B_\tau) \exp(c_\tau))$ is the unique solution of our equation but computations in Example 9.8.2 imply that in this case there is no nonnegative solution. $\square$